



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

#### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT | AI-Powered Sprint Retrospective Analysis System |                |
|----------------------|-------------------------------------------------|----------------|
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### Abstract

Agile software development relies on regular sprint retrospectives to identify challenges, evaluate team performance, and improve future development cycles. However, analyzing large amounts of team feedback manually can be time-consuming and may result in important issues being overlooked. This project proposes an AI-Powered Sprint Retrospective Analysis and Agile Improvement Recommendation System that analyzes sprint information and team feedback to identify recurring problems and provide actionable recommendations for improving the software development process.

The proposed system allows team members to submit sprint feedback along with basic sprint details such as planned tasks, completed tasks, delayed tasks, reported issues, and team performance ratings. Natural Language Processing (NLP) techniques are used to analyze textual feedback and identify important factors such as sentiment, common problems, workload issues, communication difficulties, requirement changes, and testing challenges. A dashboard can be developed to visualize sprint performance, feedback trends, identified issues, and suggested improvements.

The proposed solution can be implemented using technologies such as Python, NLP libraries, machine learning techniques, and a web-based dashboard. The system can use classification and clustering approaches to categorize feedback and detect recurring patterns. In future implementations, advanced Large Language Models (LLMs) can be integrated to understand complex feedback and generate more context-aware recommendations.

Overall, the AI-Powered Sprint Retrospective Analysis and Agile Improvement Recommendation System aims to make sprint retrospectives more efficient, objective, and data-driven. The system can help Agile teams identify recurring challenges, improve collaboration, optimize workload distribution, and continuously enhance their software development process.

Signature of the Guide

HOD